



Kernel Density Weighted Twin SVM: A Robust Approach to Noisy Data

Abstract:

We address the sensitivity of Twin Support Vector Machine variants to label noise by proposing a density-weighted Least Squares Twin Support Vector Machine (KDEWLTSVM). The method integrates kernel density estimation (KDE) into the least-squares optimization to assign global density-based sample weights, which naturally reduce the influence of low-density and potentially mislabeled samples while avoiding graph construction used in locally weighted approaches. Experiments on three binary classification datasets with simultaneous feature perturbations and 20% label noise show that KDEWLTSVM consistently outperforms LTSVM and LSWLTSVM in both accuracy and overall ranking using a linear kernel. These results demonstrate that density-based weighting improves robustness and generalization under noisy conditions.

Keywords: Twin support vector machine, Kernel density estimation, Label noise robustness, and Density-based weighting.

Mathematics Subject Classification (2010): 68T10, 62H30, 62G07.

1 Introduction

Support Vector Machines (SVMs) are a fundamental class of learning algorithms based on statistical learning theory and convex optimization. They formulate classification and regression as margin maximization problems and determine a single optimal separating hyperplane through a quadratic programming scheme. Although classical SVMs have solid theoretical foundations and practical success, their computational com-

plexity escalates with the size and dimensionality of data [Tanveer et al. \(2021\)](#); [Tian et al. \(2014\)](#). This limitation encouraged researchers to introduce new formulations aiming for better efficiency and scalability.

Twin Support Vector Machine (TSVM) marked an important improvement by constructing two non-parallel hyperplanes through the solution of two smaller quadratic programs. This decomposition decreased computational burden while preserving competitive generalization capability [Quadir et al. \(2025\)](#); [Leone et al. \(2025\)](#). Later, the Least Squares Twin Support Vector Machine (LSTSVM) was developed by converting inequality constraints into equality constraints using the least-squares principle. It replaced quadratic programs with linear equations and removed the dependence on external solvers. The resulting framework enabled fast training on large datasets and became particularly useful for real-time applications [Fongmun and Mouktonglang \(2025\)](#); [Hua et al. \(2015\)](#); [Tanveer et al. \(2016\)](#).

Nevertheless, both TSVM and LSTSVM are quite vulnerable to label noise that frequently appears in practical datasets due to annotation or measurement errors [Zhu et al. \(2023\)](#). Misclassified samples may distort the optimization process and degrade classification stability. Several extensions have been proposed to address this problem. Among them, the Least Squares Weighted Twin Support Vector Machine with Local Information (LSWLTSVM) achieved robust performance through locally weighted optimization [Hua et al. \(2015\)](#). However, its weighting mechanism is based on a local k NN adjacency graph, which relies on heuristic similarity measures and captures only neighborhood relationships rather than the underlying data distribution. Consequently, this local graph-based weighting is not well suited to handle label noise. In practice, a misclassified sample usually lies far from the true distribution of the class to which it is incorrectly assigned. Such points typically appear in low-density regions of that class, yet local graph-based weighting cannot reliably identify or suppress their influence.

To address this limitation, the present work introduces a density-aware weighted LSTSVM that integrates nonparametric density estimation into the optimization framework. Specifically, kernel density estimation (KDE) is employed to estimate the local probability density around each sample and to derive adaptive weights. Samples located in low-density regions, which are more likely to correspond to misclassified or atypical observations, naturally receive smaller weights. As a result, their influence on the construction of the hyperplanes is reduced. This reduction enhances robustness against label noise and simultaneously avoids explicit graph construction. The proposed KDE-weighted LSTSVM (KDEWLSTSVM), preserves the computational efficiency of least-squares learning while establishing a statistically principled mechanism for enhancing robustness and generalization in noisy data environments. [Tanveer et al. \(2021\)](#).

2 The LSTSVM

In this section, we present a summary of the LSTSVM. The LSTSVM is developed on the basis of the TSVM by using equality constraints instead of inequality constraints and also, the optimization problem degenerates into the optimal solution problem of two linear equation systems. Suppose we have a training set

$$D = \{(x_i, y_i) | x_i \in \mathbb{R}^d, y_i \in \{1, -1\}, i = 1, 2, \dots, n\}.$$

By dividing the training set into m_1 positive class sample points and m_2 negative class sample points, A and B denote the data matrices of the positive and negative classes, respectively. Then the LSTSVM is defined as follows

$$\begin{aligned} \min_{\mathbf{w}_1, b_1} \quad & \frac{1}{2} \|A\mathbf{w}_1 + \mathbf{e}_1 b_1\|^2 + \frac{c_1}{2} \boldsymbol{\xi}_1^\top \boldsymbol{\xi}_1 \\ \text{s.t.} \quad & -(B\mathbf{w}_1 + \mathbf{e}_2 b_1) + \boldsymbol{\xi}_1 = \mathbf{e}_2, \boldsymbol{\xi}_1 \geq 0, \end{aligned} \quad (2.1)$$

and

$$\begin{aligned} \min_{\mathbf{w}_2, b_2} \quad & \frac{1}{2} \|B\mathbf{w}_2 + \mathbf{e}_2 b_2\|^2 + \frac{c_2}{2} \boldsymbol{\xi}_2^\top \boldsymbol{\xi}_2 \\ \text{s.t.} \quad & (A\mathbf{w}_2 + \mathbf{e}_1 b_2) + \boldsymbol{\xi}_2 = \mathbf{e}_1, \boldsymbol{\xi}_2 \geq 0, \end{aligned} \quad (2.2)$$

where $\mathbf{w}_1, \mathbf{w}_2 \in \mathbb{R}^d$ are the weight vectors, the biases are $b_1, b_2 \in \mathbb{R}$, $c_1, c_2 > 0$ are penalty parameters, $\boldsymbol{\xi}_1, \boldsymbol{\xi}_2$ are slack variables with dimension $m_2 \times 1$ and $m_1 \times 1$ respectively, $\mathbf{e}_1, \mathbf{e}_2$ are the column vectors of appropriate dimensionality with all 1 components.

By substituting the equality constraints of quadratic programming problem (QPP) (2.1) into the objective function, setting the gradient of QPP with respect to $\mathbf{w}_1$ and b_1 to zero and arranging them in matrix form, solving for $\mathbf{w}_1$ and b_1 and defining $E = [A \quad \mathbf{e}_1]$ and $F = [B \quad \mathbf{e}_2]$ we have

$$[\mathbf{w}_1 \quad b_1]^\top = -(F^\top F + \frac{1}{c_1} E^\top E)^{-1} F^\top \mathbf{e}_2.$$

In a similar way the solution of QPP (2.2) can be calculate as

$$[\mathbf{w}_2 \quad b_2]^\top = (E^\top E + \frac{1}{c_2} F^\top F)^{-1} E^\top \mathbf{e}_1.$$

For the nonlinear case, the kernel trick is employed to map the data into a high-dimensional feature space, and the resulting formulations are obtained in a manner similar to the linear LSTSVM.

3 The KDEWLSTSV

Following [Hua et al. \(2015\)](#), define the coefficients of the LSWLTSVM as

$$[\mathbf{w}_1 \quad b_1]^\top = - \left[F^\top F + \frac{1}{c_1} E^\top D^{(1)} E \right]^{-1} F^\top D^{(2)} \mathbf{e}_2,$$

and

$$[\mathbf{w}_2 \quad b_2]^\top = \left[E^\top E + \frac{1}{c_2} F^\top D^{(2)} F \right]^{-1} E^\top D^{(1)} \mathbf{e}_1,$$

where

$$D^{(C)} = \text{diag}(\pi_1^{(C)}, \dots, \pi_{m_C}^{(C)}), \quad C \in \{1, 2\},$$

is the diagonal weight matrix whose elements $\pi_i^{(C)}$ are obtained from a local k NN adjacency graph. Specifically, each weight is computed according to a hot kernel similarity between neighboring samples, and is nonzero only if the corresponding samples belong to each other's k nearest neighbors. Hence, the weighting scheme reflects local neighborhood relationships rather than a global estimation of the underlying data distribution. In our approach, $\pi_i^{(C)}$ is computed from the KDE. Given class- C samples $\{x_i^{(C)}\}_{i=1}^{m_C} \subset \mathbb{R}^d$, the KDE of the density function $p(x | C)$ is

$$\begin{aligned} \pi_i^{(C)} &= \hat{p}(x | C) \\ &= \frac{1}{m_C h_C^d} \sum_{i=1}^{m_C} K\left(\frac{x - x_i^{(C)}}{h_C}\right), \end{aligned} \quad (3.1)$$

where $K(\cdot)$ is a kernel satisfying $\int K(u) du = 1$ and $h_C > 0$ is the bandwidth. Our approach provides a global density-based weighting that naturally downweights low-density and potentially mislabeled samples. In this paper, we use the Gaussian kernel,

$$K(u) = (2\pi)^{-\frac{d}{2}} \exp\left(-\frac{1}{2}\|u\|^2\right).$$

4 Experimental Results

In this section, we conduct an extensive experimental study to empirically assess the performance of the KDEWLSTSV. Given that the central contribution of this work concerns the explicit modeling of label noise, KDEWLSTSV is primarily compared with LSTSV and LSWLTSVM. Experiments are carried out using a linear kernel.

To ensure fair evaluation, hyperparameters of all methods are selected via a parallelized grid-search procedure with five-fold cross-validation on the training set. For each classifier, the regularization parameters are set equal ($c_1 = c_2$) and searched over $\{10^{-4}, 10^{-3}, \dots, 10^3\}$. The KDEWLSTSV uses

Gaussian kernel density estimation, where the bandwidth parameter is tuned over $\{0.1, 0.3, 0.5, \dots, 1.9\}$. For LSWLTSVM, the neighborhood size k is chosen from $\{1, 2, \dots, 8\}$ and the hot kernel exponent t is searched over $\{0.5, 1, 2, 4, 8, 16, 32\}$.

Classification accuracy is evaluated on noise-free test sets to ensure an unbiased assessment of predictive performance. In addition to accuracy, robustness is further examined by ranking the competing classifiers based on their overall performance. Higher ranks are interpreted as indicative of greater resilience to both label noise and feature perturbations.

The experimental evaluation is performed on three binary classification datasets drawn from the LibSVM repository (Chang and Lin, 2001) and the UCI Machine Learning Repository (Bache and Lichman, 2013). Prior to training, all features are scaled to the interval $[0, 1]$. The selected datasets exhibit substantial diversity in terms of sample size, ranging from 145 to 768 instances, thereby facilitating a comprehensive and representative comparison of the evaluated methods. To maintain consistency across experiments, only binary classification problems are considered. All datasets are carefully preprocessed before analysis, and any instances with missing values are removed to prevent potential biases and to ensure the integrity of the experimental results. Detailed characteristics of the datasets are summarized in Table 1.

Table 1: Characteristics of the benchmark datasets.

Dataset	Number of Samples	Number of Features
Liver disorders	145	5
Breast cancer wisconsin diagnostic	569	30
Pima Indians Diabetes	768	8

Table 2: Comparison of classification accuracy (%) and ranking using **Linear Kernel** under feature noise ($\sigma = 0.2$) and 20% label noise.

Dataset	LSTSVM	LSWLTSVM	KDEWLTSVM
Liver Disorders	72.41 (2.5)	72.41 (2.5)	75.86 (1)
Breast Cancer Wisconsin Diagnostic	85.40 (3)	95.62 (2)	97.08 (1)
Pima Indians Diabetes	76.62 (3)	77.27 (2)	77.92 (1)
Final Rank	2.83	2.17	1
Accuracy Mean	78.14	81.77	83.62

4.1 Results

Table 2 summarizes the classification accuracy and ranking of the compared methods under 20% label noise using a linear kernel. In addition, 40% of the feature columns were selected, and within each selected

column, 40% of the training samples were corrupted by additive zero-mean Gaussian noise with a standard deviation of $\sigma = 0.2$. As observed, the KDEWLSTSVm consistently achieves the highest accuracy across all datasets and outperforms both the LSTSVm and the LSWLSTSVm. In particular, the KDEWLSTSVm attains notable improvements on the Breast Cancer Wisconsin Diagnostic and Liver Disorders datasets, demonstrating superior robustness to simultaneous feature and label noise. The overall mean accuracy and final ranking also confirm its leading performance (mean accuracy 83.62%, rank 1). These results indicate that the density-based weighting effectively improves generalization under noisy conditions.

Discussion and Conclusion

The KDEWLSTSVm used KDE-based global density weighting within the LSTSVm and outperformed the LSTSVm and the LSWLSTSVm under 20% label noise. Results were limited to three small linear-kernel binary datasets without significance or sensitivity analyses. Future work should test larger datasets and extend to nonlinear and multiclass settings with careful bandwidth and efficiency studies.

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